MINDSCREEN

AI-Powered Burnout Detection System

Hackathon Project Report — Summer Hack 2026

Event	Summer Hack 2026
Project	MindScreen
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Live Demo	https://mindscreen-one.vercel.app/
Repository	github.com/yashtambade56-ux/Mindscreen

1. Executive Summary

MindScreen is a browser-based mental health and burnout monitoring dashboard built for Summer Hack 2026. It fuses wearable fitness data, webcam-based facial emotion detection, and passive behavioral signals (typing speed, mouse activity) into a single composite Burnout Score, and layers an in-context AI assistant, AXON v3, on top to translate that score into plain-language, tactical advice. The interface is styled as a cyberpunk HUD, giving the project a distinctive, demo-friendly identity rather than a generic dashboard look.

2. Problem Statement

Students and working professionals increasingly face stress and burnout, but most existing wellness tools share the same three gaps:

- They rely on manual self-reported check-ins rather than continuous signals.
- They don't track state in real time, so warning signs are caught late.
- Feedback is generic ("take a break") instead of personalized to the user's current data.

MindScreen's goal is to replace manual, occasional self-assessment with continuous, passive, and intelligent monitoring that a user can glance at throughout the day.

3. Solution Overview

The application continuously (or on-demand, in the current build) reads four categories of signal and combines them into a single 0–100% Burnout Score:

- **Physical activity:** steps and calories from Google Fit
- **Emotional state:** facial expression detected via webcam
- **Behavioral signals:** typing speed (WPM) and mouse-movement intensity captured passively in the browser

- **Vital stats:** heart rate (BPM) and SpO2, shown with an animated ECG-style pulse visualization

The score drives a three-tier status system (Optimal / Warning / Critical) that changes the UI's color language and copy, and is also fed into AXON v3 as context so its advice reflects the user's actual current state rather than a generic script.

4. Core Features

4.1 Google Fit Integration

- OAuth 2.0 authentication against a user's Google account.
- Fetches step count and calories via the fitness/v1/users/me/dataset:aggregate endpoint, reading the com.google.step_count.delta and com.google.calories.expended data streams.
- Compatible with data synced from smartwatches and fitness bands that write into Google Fit.

4.2 AXON v3 AI Assistant

- Backed by Groq's hosted Llama 3.1 8B Instant model over its OpenAI-compatible chat completions API.
- Given a system prompt that fixes its persona ("tactical, slightly robotic but empathetic") and injects the user's live mood, burnout score, heart rate, and step count so every reply is grounded in current data.
- Exposes quick commands: BIO_REPORT, STRESS_ANALYSIS, TACTICAL_ADVICE, and SYSTEM_SCAN.
- Responses are capped at roughly 60 words to keep the HUD aesthetic snappy.

4.3 Facial Mood Detection

- Uses face-api.js (TinyFaceDetector + expression model) running client-side against the webcam feed.
- Classifies expression into happy, sad, angry, neutral, or surprised, each with a confidence score.
- A high-confidence "surprised" reading automatically triggers the app's Boost Mode animation, used as a proxy for a sudden energy spike.

4.4 Real-Time Health Monitoring

- Displays heart rate (BPM) and blood-oxygen level (SpO2) alongside an animated ECG-style pulse component.
- In the current build these vitals are simulated / demo-seeded rather than read from a connected biosensor, since no wearable heart-rate API was integrated during the hackathon.

4.5 Passive Activity Tracking

- A custom hook listens to window-level mousemove and keydown events.
- Every 5 seconds it recomputes typing speed (characters typed ÷ 5, normalized to a per-minute rate) and a raw mouse-movement count, then resets the counters — giving a lightweight, non-invasive read on how actively the user is working.

4.6 Burnout Score & Status System

- A single 0–100% score drives three status bands: Optimal (blue), Warning (yellow, roughly score > 50), and Critical (red, roughly score > 80), each with its own advice string and avatar art.
- A demo-profile switcher lets judges instantly load five preset personas (from a fully rested "Optimal" user to a "Critical" burnout case) to showcase the full range of the UI without waiting on real sensor data.

4.7 Voice Feedback

- Uses the browser's Web Speech API to read AXON's replies aloud in a synthesized voice, reinforcing the "AI system" feel of the HUD.

4.8 Boost Mode

- A triggerable high-energy state with its own animation and system-wide visual treatment, fired either manually or automatically from a strong "surprised" mood detection.

4.9 UI / Design

- Cyberpunk HUD aesthetic with cyan/rose neon accents, an X-ray-style avatar panel, and a fully responsive layout built with Tailwind CSS and custom CSS.

5. Tech Stack

Frontend	React 19 + Vite 8
Styling	Tailwind CSS 4 + custom CSS (Grid, Flexbox, neon UI)
AI Model	Groq API — Llama 3.1 8B Instant
Face Detection	face-api.js (TinyFaceDetector + expression model)
On-device ML	TensorFlow.js + coco-ssd (object detection utilities)
Wearables API	Google Fit REST API (OAuth 2.0)
Voice	Web Speech API
Animation	anime.js
Deployment	Vercel
Version Control	Git & GitHub

6. System Architecture & Working Flow

The app is a single-page React application; each subsystem is isolated into its own service or component and composed together in App.jsx, which owns the shared state (score, mood, vitals, chat log) that every panel reads from.

1. User connects Google Fit via OAuth 2.0 (googleFit.js), or loads a preset demo profile for an instant walkthrough.

2. Steps and calories are fetched from the Fit API; the webcam starts streaming into face-api.js for mood detection.

3. Keyboard and mouse listeners passively accumulate activity, recomputing typing speed and movement intensity every 5 seconds.

4. These signals combine into the Burnout Score, which sets the Optimal / Warning / Critical status and updates the HUD's color language and avatar art.

5. The current mood, score, heart rate, and step count are packaged into AXON v3's system prompt and sent to Groq's Llama 3.1 endpoint on each chat turn.

6. AXON's reply is displayed in the chat panel and optionally read aloud through the Web Speech API.

7. Target Users & Impact

Target Users

- Students
- Software developers
- Working professionals
- Fitness-tracker users
- Smartwatch companies (as a potential white-label dashboard)

Impact

- Helps users notice early signs of burnout instead of realizing it after the fact.
- Builds general productivity and self-awareness around work habits.
- Encourages small, timely interventions rather than one-off wellness check-ins.

8. Limitations

- MindScreen is a wellness prototype, not a medical or diagnostic tool.
- Facial-emotion accuracy depends on webcam quality, lighting, and camera angle.
- Google Fit integration requires the user to have fitness data already syncing from a device or app.
- Heart rate and SpO2 are currently simulated for demo purposes rather than pulled from a connected biosensor.
- The Burnout Score is a heuristic estimate, not a clinically validated measurement.

9. Future Scope

- Sleep tracking integration.
- Multi-device support: Apple Health, Fitbit, Garmin.
- A historical analytics dashboard for long-term trend tracking.
- Firebase authentication for persistent, multi-user accounts.

- A React Native mobile companion app.
- A guided meditation and breathing assistant built into AXON.
- Replacing simulated heart-rate/SpO2 with a real biosensor or smartwatch data feed.

10. Conclusion

Within the scope of Summer Hack 2026, the team shipped an end-to-end, demo-ready prototype that pulls together wearable data, computer-vision-based emotion detection, passive behavioral tracking, and an LLM-driven assistant behind a single, cohesive Burnout Score. The strongest parts of the submission are the breadth of signal sources it fuses and the distinctive HUD presentation; the clearest next steps are replacing the simulated vitals with real sensor data and validating the scoring heuristic against real user feedback.

11. Reference Links

- Live demo: https://mindscreen-one.vercel.app/
- Hackathon report (Notion): notion.so — "MindScreen Hackathon Report"
- Demo video: Google Drive link (see repository README)
- Repository: github.com/yashtambade56-ux/Mindscreen